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Roll No. :

FIRST PRE-BOARD EXAM. (2024-2025)

SUBJECT : BIOLOGY

CLASS : XII

Time : 3 hours.

Marks : 70

General Instructions :

- 1. All questions are compulsory.***
- 2. The question paper has five sections and 33 questions. All questions are compulsory.***
- 3. Section–A has 16 questions of 1 mark each; Section–B has 5 questions of 2 marks each; Section– C has 7 questions of 3 marks each; Section– D has 2 case-based questions of 4 marks each; and Section–E has 3 questions of 5 marks each.***
- 4. There is no overall choice. However, internal choices have been provided in some questions. A student has to attempt only one of the alternatives in such questions.***
- 5. Wherever necessary, neat and properly labeled diagrams should be drawn.***

SECTION-A

1. Pyramid of energy is : 1
 - a) Always upright
 - b) Both Always inverted and Always upright
 - c) Sometimes inverted sometimes upright
 - d) Always inverted

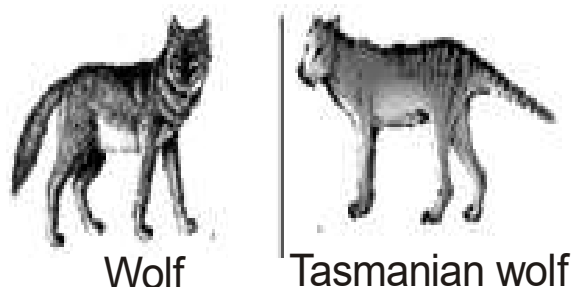
2. Which of the following birth control measure can be considered as the safest? 1
- a) Termination of unwanted pregnancy
 - b) The rhythm method
 - c) The use of physical barriers
 - d) Sterilization techniques
3. The multipurpose protected area which is meant for preserving genetic diversity in the representative ecosystem is called _____.1
- a) National parks
 - b) Protected areas
 - c) Biosphere reserves
 - d) Sanctuaries
4. In a population, the condition at which the rate of addition of new members is more than the rate of individuals lost indicates : 1
- a) Zero population growth
 - b) Fluctuating growth
 - c) Exponential growth
 - d) Declining growth
5. The bp of DNA double helix is given below. Suitable m RNA strand that is derived from transcription of 3' ATTTCC 5' is 5' TAAAGG3':1
- a) UAAAGG
 - b) CUUUCC
 - c) GAAAGG
 - d) CCUUUC
6. Which of the following is now being commercially produced by biotechnological procedures? 1

- | | |
|-------------|------------|
| a) Nicotine | b) Insulin |
| c) Morphine | d) Quinine |

7. A woman with normal vision, but whose father was colour-blind, marries a colour-blind man. Suppose that the fourth child of this couple was a boy. This boy : 1

- a) May be colour-blind or may be of normal vision
- b) Must be colour-blind
- c) Will be partially colour-blind since he is heterozygous for the colour blind mutant allele
- d) Must have normal colour vision

8. Picture shown below is an example of : 1

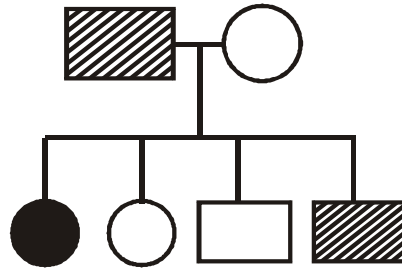


- (a) Convergent evolution of Australian marsupials and placental mammals
- (b) Divergent evolution of Australian marsupials and placental mammals
- (c) Homologous organs of both animals
- (d) Analogous organs of both animals.

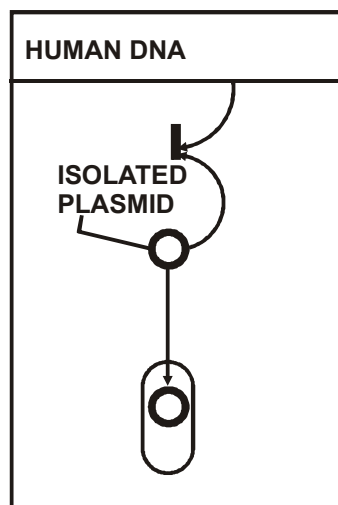
9. During food chain the maximum energy is stored in : 1

- | | |
|----------------|---------------|
| a) Carnivores | b) Producers |
| c) Decomposers | d) Herbivores |

10. Study the pedigree chart of a certain family given below and select the correct conclusion which can be drawn for the character : 1



- (a) The trait under study could not be colour blindness
 (b) The female parent is heterozygous
 (c) The male parent is homozygous dominant
 (d) The parent could not have had a normal daughter for this character
11. Monascus purpureus is a yeast commercially used in the production of : [1]
 a) Citric acid b) Ethanol
 c) Statins d) Streptokinase
12. Arrange the following steps that are shown in the figure : 1



1. The plasmid is taken up into a bacterial cell which makes protein directed by human DNA
2. DNA segment incorporated into the bacterial plasmid
3. The segment of DNA removed from human cell
4. In Genetic engineering (Recombinant DNA technology)

- a) $1 \rightarrow 4 \rightarrow 3 \rightarrow 2$
- b) $4 \rightarrow 3 \rightarrow 2 \rightarrow 1$
- c) $1 \rightarrow 2 \rightarrow 3 \rightarrow 4$
- d) $2 \rightarrow 3 \rightarrow 4 \rightarrow 1$

13. Assertion (A) : Condoms are barriers made of thin rubber/latex sheath and their use is increased in recent years. 1

Reason (R) : It prevents conception and its additional benefit is protecting the user from contracting STDs and AIDS.

- a) Both A and R are true and R is the correct explanation of A.
- b) Both A and R are true but R is not correct explanation of A.
- c) A is true but R is false.
- d) A is false but R is true.

14. Assertion: Different type of alcoholic drinks depending on the type of raw material used for fermentation. 1

Reason : With or without distillation different types of alcoholic drinks are obtained.

- a) Assertion and reason both are correct statements and reason is correct explanation for assertion.
- b) Assertion and reason both are correct statements but reason is not correct explanation for assertion.

- c) Assertion is correct statement but reason is wrong statement.
- d) Assertion is wrong statement but reason is correct statement.
15. Assertion (A): Net primary productivity is always less than Gross primary productivity in an ecosystem. 1
- Reason (R) : A considerable amount of Gross primary productivity is utilized by plants in respiration.
- a) Both A and R are true and R is the correct explanation of A.
- b) Both A and R are true but R is not correct explanation of A.
- c) A is true but R is false.
- d) A is false but R is true.
16. **Assertion (A):** Ramapithecus was more man like while Dryopithecus was more ape like. 1
- Reason (R):** These primates were hairy and walked like gorillas and chimpanzees.
- a) Both A and R are true and R is the correct explanation of A.
- b) Both A and R are true but R is not correct explanation of A
- c) A is true but R is false.
- d) A is false but R is true.

SECTION-B

17. What was the specialty of the milk produced by the transgenic cow Rosie?

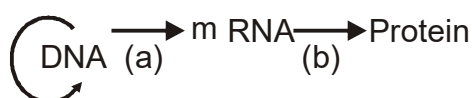
OR

How are GM plants better than normally grown plants? (any 4 points)2

18. The flow of genetic information is shown below. Name the process

of (a) and (b).

2



19. In the diagram given below, show the path of a pollen tube from the pollen on the stigma into the embryo sac. Name the components of egg apparatus.

2



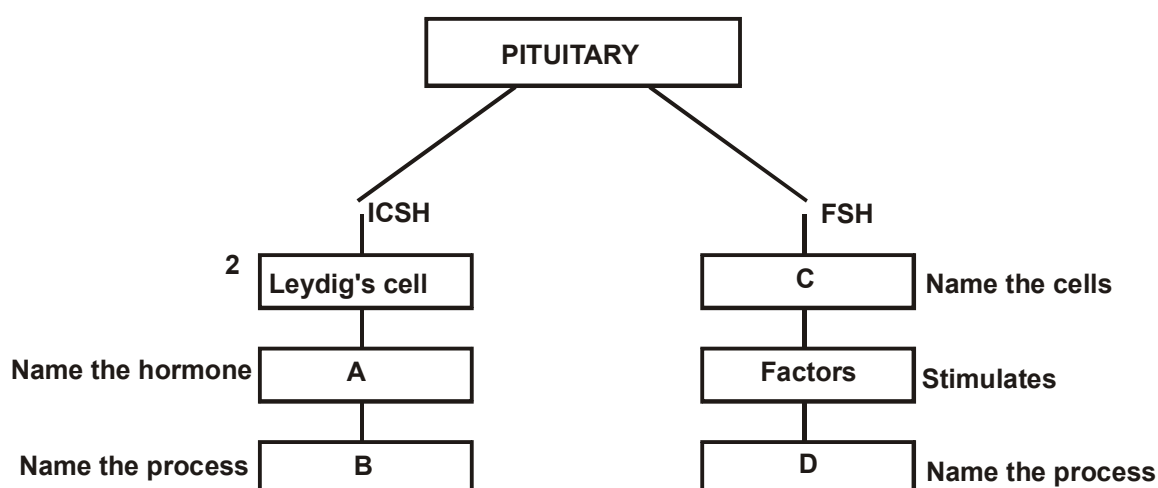
OR

Write four characteristics of an insect pollinated flower.

2

20. Given below is an incomplete chart showing the influence of hormones on gametogenesis in males? Observe the chart carefully and fill in the blanks A, B, C and D.

2



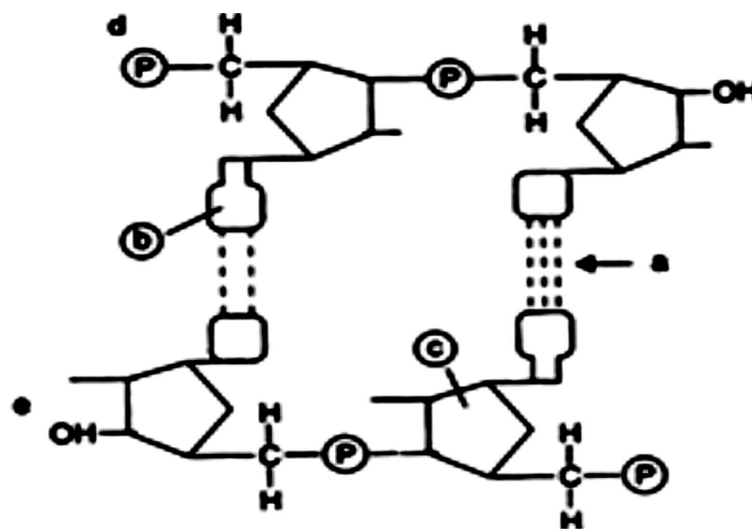
21. Give two examples of natural predators of insect-pests.

OR

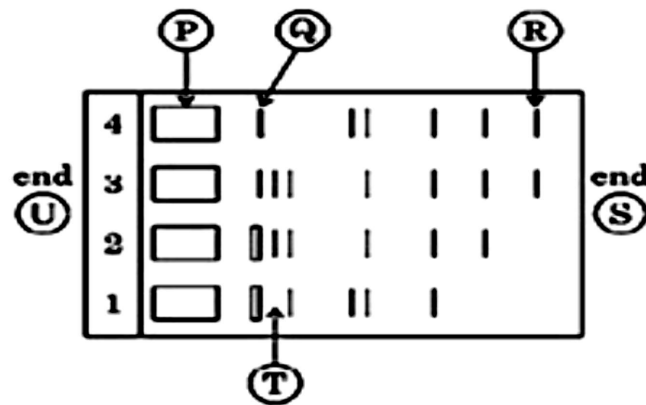
Bottled fruit juices are clearer as compared to those made at home. Explain.

SECTION-C

22. Study the given portion of double-stranded polynucleotide chain carefully. Identify a, b, c and d. 3



23. How the alleles of particular genes differ from each other? Explain its significance. [3]
24. Define the following terms and give one example for each: [3]
- | | |
|------------------------------|---------------|
| a. Commensalism | b. Parasitism |
| c. Camouflage | d. Mutualism |
| e. Interspecific competition | |
25. (a) Given below is the stepwise schematic representation of the process of electrophoresis. Identify the **alphabets** representing (i) Anode end (ii) smallest/lightest DNA strand in the matrix (iii) Agarose gel



b. What is elution? State the importance of elution in this process.

26. Write short notes on the following : 3

- i. Wildlife sanctuary ii. Biosphere reserves

27. Define the terms [3]

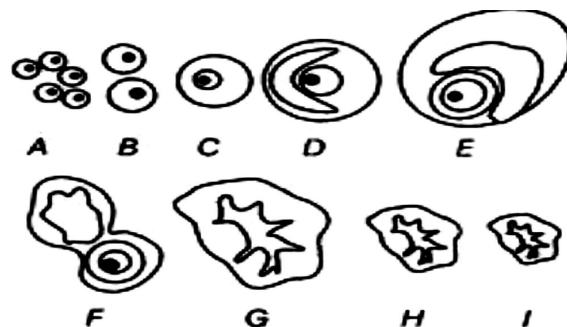
- i. Palaeontology ii. Phylogeny iii. Fossils

28. Prevention is better than cure is an apt slogan to safeguard adolescents from drug abuse. List any 6 steps that could be taken in this regard. 3

SECTION-D

29. Read the following text carefully and answer the questions that follow :

The following is the illustration of the sequence of ovarian events (A-I) in a human female. 4



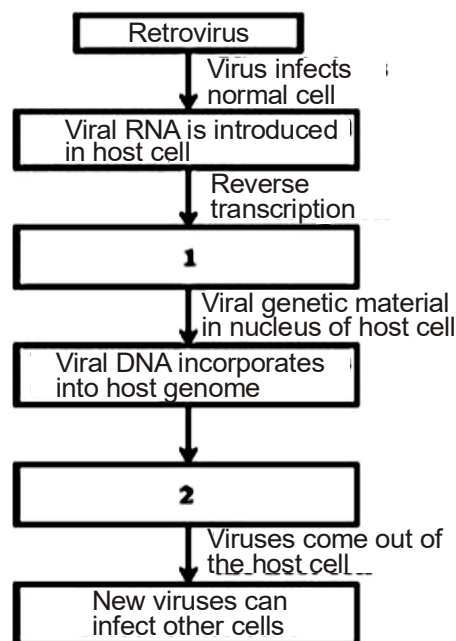
- i. Identify the figure that illustrates ovulation and mention the stage of menstrual it represents. (1)
- ii. Name the ovarian hormone and the pituitary hormone that have caused the above mentioned event. (1)
- iii. Explain the changes that occur in the uterus simultaneously in anticipation. (2)

OR

Write the differences between C and H. (2)

30. Read the following text carefully and answer the questions that follow : 4

The diagram shows the replication of the retrovirus in the host.



- (i) Fill in the missing data in boxes labelled 1 & 2. (1)
- (ii) Why is it named as retrovirus? (1)
- (iii) While the virus is being replicated and released, does the infected cell survive and why the virus infected cells prevent spreading of virus to healthy cells? (2)

OR

What is the effect of HIV infection on immune system? (2)

SECTION-E

31. Draw a labelled sketch of L.S. of pistil showing the process of double fertilization. 5

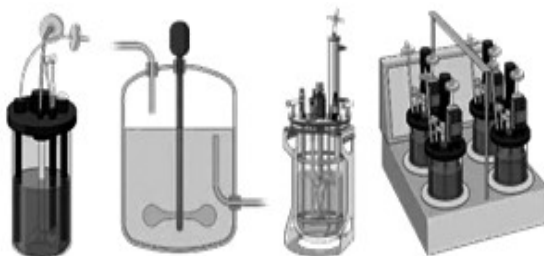
OR

Draw a diagram of male gametophyte of an angiosperm. Label any four parts. Why sporopollenin is considered the most resistant organic material?

32. Define an operon. Giving an example, explain an Inducible operon. 5
OR

Name and describe the steps involved in the technique widely used in forensics that serves as the basis of paternity testing in case of disputes.

33. Bioreactors are vessels that have been designed and produced to provide an effective environment for enzymes or whole cells to transform biochemicals into products. In some cases, inactivation of cells or sterilization is carried out in the bioreactor such as in water treatment. The development of bioreactors is required to produce large quantities of products. 5



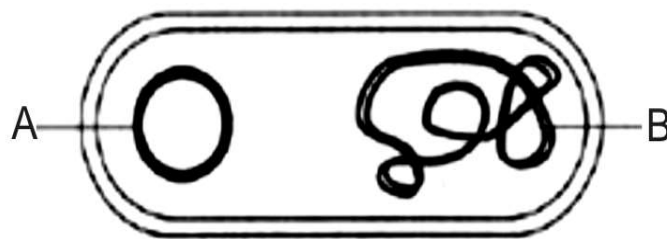
- (i) Give optimum growth conditions used in bioreactors.
ii. Draw a well labelled diagram of simple stirred-tank bioreactor.

- iii. How does a simple stirred tank bioreactor differ from sparged stirred tank bioreactor ?

OR

Recombinant DNA (rDNA) is a technology that uses enzymes to cut and paste together DNA sequences of interest. The recombined DNA sequences can be placed into vehicles called vectors that ferry the DNA into a suitable host cell where it can be copied or expressed.

- i. A bacterial cell is shown in the figure given below. Label the part 'A' and 'B'. Also, mention the use of part A in rDNA technology.



- ii. Suppose a linear DNA fragment and a plasmid has three restriction sites for EcoRI. How many fragments will be produced from linear DNA and plasmid, respectively?